

AIAC ASSIGNMENT – 4

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BATCH : 32

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Q1. Zero-Shot Prompting (Basic Lab Task)

Task:

Write a Python function that classifies a given text as Spam or Not Spam using zero-shot prompting.

code :

```
Lab_Ass_4 > Task1.py > ...
1  #Write a Python function that classifies a given text as Spam or Not Spam using zero-shot prompting
2
3  def classify_text(text):
4      # Zero-shot prompting approach
5      prompt = f"Classify the following text as Spam or Not Spam: {text}"
6      # In a real implementation, you would use an LLM API here
7      # For now, we'll simulate the classification
8      if "free" in text.lower() or "won" in text.lower():
9          return "Spam"
10     else:
11         return "Not Spam"
12
13     text = "Congratulations! You have won a free lottery ticket."
14     result = classify_text(text)
15     print(result)
```

output :

```
PS D:\courses\AIAC> & C:/Users/ANJANI/AppData/Local/Programs/Python/Python313/python.exe d:/courses/AIAC/Lab_Ass_4/Task1.py
Spam
```

Q2. One-Shot Prompting (Emotion detection)

Task:

Write a Python program that detects the emotion of a sentence using one-shot prompting.

Emotions: ['happy', 'sad', 'angry', 'excited', 'nervous', 'neutral']

code :

```
Lab_Ass_4 > Task2.py > ...
1  '''
2  Emotions: ['happy', 'sad', 'angry', 'excited', 'nervous', 'neutral']
3  sentence : "I got selected for my dream job!"
4  output : excited
5  '''
6  sentence = input("Enter a sentence: ")
7
8  if "happy" in sentence:
9      | print("happy")
10 elif "sad" in sentence:
11     | print("sad")
12 elif "angry" in sentence:
13     | print("angry")
14 elif "excited" in sentence:
15     | print("excited")
16 elif "nervous" in sentence:
17     | print("nervous")
18 elif "neutral" in sentence:
19     | print("neutral")
20 else:
21     | print("Emotion not recognized")
22
```

output :

```
PS D:\courses\AIAC> & C:/Users/ANJANI/AppData/Local/Programs/Python/Python313/python.exe d:/courses/AIAC/Lab_Ass_4/Task2.py
Enter a sentence: I am very happy for your success .
happy
```

Q3. Few-Shot Prompting (Student Grading Based on Marks)

Task:

Write a Python program that predicts a student's grade based on marks using few-shot prompting.

Grades:

['A', 'B', 'C', 'D', 'F']

code :

```
Lab_Ass_4 > Task3.py > ...
1  '''
2  prediction of student grading
3  Marks: 95 display Grade: A
4  Marks: 82 display Grade: B
5  Marks: 74 display Grade: C
6  Marks: 65 display Grade: D
7  Marks: 45 display Grade: F
8  '''
9
10 marks = int(input("Enter the marks: "))
11
12 if marks >= 95:
13     print("Grade: A")
14 elif marks >= 82:
15     print("Grade: B")
16 elif marks >= 74:
17     print("Grade: C")
18 elif marks >= 65:
19     print("Grade: D")
20 else:
21     print("Grade: F")
```

output :

```
PS D:\courses\AIAC> & C:/Users/ANJANI/AppData/Local/Programs/Python/Python313/python.exe d:/courses/AIAC/Lab_Ass_4/Task3.py
Enter the marks: 79
Grade: C
```

Q4. Multi-Shot Prompting (Indian Zodiac Sign Prediction using Month Name)

Task:

Write a Python program that predicts a person's Indian Zodiac sign (Rashi) based on the month of birth (month name) using multi-shot prompting.

code :

```
Lab_Ass_4 > Task4.py > ...
1  '''
2  prediction of indian zodiac sign
3  examples:
4  March - Display as Mesha
5  April - Display as Vrishabha
6  May - Display as Mithuna
7  June - Display as Karka
8  July - Display as Simha
9  August - Display as Kanya
10 September - Display as Tula
11 October - Display as Vrischika
12 November - Display as Dhanu
13 December - Display as Makara
14 January - Display as Kumbha
15 February - Display as Meena
16 '''
17 month = input("Enter the month: ")
18 if month == "March":
19     | print("Mesha")
20 elif month == "April":
21     | print("Vrishabha")
22 elif month == "May":
23     | print("Mithuna")
24 elif month == "June":
25     | print("Karka")
26 elif month == "July":
27     | print("Simha")
28 elif month == "August":
29     | print("Kanya")
30 elif month == "September":
31     | print("Tula")
32 elif month == "October":
33     | print("Vrischika")
34 elif month == "November":
35     | print("Dhanu")
36 elif month == "December":
37     | print("Makara")
38 elif month == "January":
39     | print("Kumbha")
40 elif month == "February":
41     | print("Meena")
42 else:
43     | print("Invalid month")
```

output :

```
PS D:\courses\AIAC> & C:/Users/ANJANI/AppData/Local/Programs/Python/Python313/python.exe d:/courses/AIAC/Lab_Ass_4/Task4.py
PS D:\courses\AIAC> & C:/Users/ANJANI/AppData/Local/Programs/Python/Python313/python.exe d:/courses/AIAC/Lab_Ass_4/Task4.py
Enter the month: May
Mithuna
```

Q5. Result Analysis Based on Marks

Task: Write a Python program that determines whether a student

Passes or Fails based on marks using Chain-of-Thought (CoT)

prompting.

Result Categories:

['Pass', 'Fail']

code :

```
Task5.py X
Lab_Ass_4 > Task5.py > ...
1  '''
2  read the marks from user
3  if marks > = 40 then display as pass
4  otherwise display as fail
5  well commented code
6  '''
7  # Read the marks from user
8  marks = int(input("Enter the marks: "))
9  # Check if the marks are greater than or equal to 40
10 if marks >= 40:
11     print("Pass") # Display Pass
12 else:
13     print("Fail") # Display Fail
```

output :

```
PS D:\courses\AIAC> & C:/Users/ANJANI/AppData/Local/Programs/Python/Python313/python.exe d:/courses/AIAC/Lab_Ass_4/Task5.py
Enter the marks: 56
Pass
PS D:\courses\AIAC> & C:/Users/ANJANI/AppData/Local/Programs/Python/Python313/python.exe d:/courses/AIAC/Lab_Ass_4/Task5.py
Enter the marks: 23
Fail
```

Q6 Voting Eligibility Check (Chain-of-Thought Prompting)

Task: Write a Python program that determines whether a person is eligible to vote using Chain-of-Thought (CoT) prompting.

code :

```
Lab_Ass_4 > Task6.py > ...
1  '''
2  well commented code
3  read the age from the user
4  read there nationality then
5  check whether they are eligible to vote or not
6  if age >= 18 then display as eligible to vote
7  otherwise display as not eligible to vote
8  '''
9  age = int(input("Enter your age: "))
10 nationality = input("Enter your nationality: ")
11 if age >= 18 and nationality == "Indian":
12     print("You are eligible to vote.")
13 else:
14     print("You are not eligible to vote.")
15
```

output :

```
PS D:\courses\AIAC> & C:/Users/ANJANI/AppData/Local/Programs/Python/Python313/python.exe d:/courses/AIAC/Lab_Ass_4/Task6.py
Enter your age: 20
Enter your nationality: Indian
You are eligible to vote.
```

Q7 Prompt Chaining (String Processing – Palindrome Names)

Task: Write a Python program that uses the prompt chaining technique to identify palindrome names from a list of student names.

code :

```
Task5.py Task6.py Task7.py X
Lab_Ass_4 > Task7.py > ...
1  '''
2  generate a list of student names
3  from the above list traverse and check for the palindrome names
4  print the palindrome names
5  '''
6
7  student_names = ["Alice", "Bob", "Charlie", "David", "Eve", "Anna", "Level"]
8
9  palindrome_names = []
10
11 for name in student_names:
12     cleaned_name = ''.join(c.lower() for c in name if c.isalnum())
13     if cleaned_name == cleaned_name[::-1]:
14         palindrome_names.append(name)
15
16 print("Palindrome names:", palindrome_names)
17
```

output:

```
PS D:\courses\AIAC> & C:/Users/ANJANI/AppData/Local/Programs/Python/Python313/python.exe d:/courses/AIAC/Lab_Ass_4/Task7.py
Palindrome names: ['Bob', 'Eve', 'Anna', 'Level']
PS D:\courses\AIAC>
```

Q8 - Prompt Chaining (String Processing – Word Length

Analysis)

Task: Write a Python program that uses prompt chaining to

analyze a list of words. In the first prompt, generate a list of words.

In the second prompt, traverse the list and calculate the length of each word. In the third prompt, use the output of the previous step to determine whether each word is Short (length less than 5) or Long (length greater than or equal to 5), and display the result for each word.

code:

```
Lab_Ass_4 > Task8.py > ...
1  """
2  generate a list of words.
3  Then traverse the list and calculate the length of each word.
4  determine the word as short if its length is less than 5 or Long if its length if greater than or equal to 5.
5  """
6
7  words = ["apple", "banana", "cherry", "date", "elderberry"]
8
9  for word in words:
10     length = len(word)
11     if length < 5:
12         print(f"{word} is short")
13     else:
14         print(f"{word} is long")
```

output:

```
PS D:\courses\AIAC> & C:/Users/ANJANI/AppData/Local/Programs/Python/Python313/python.exe d:/courses/AIAC/Lab_Ass_4/Task8.py
apple is long
banana is long
cherry is long
date is short
elderberry is long
```